

Program 8:

Create Kubernetes Pods using YAML Descriptors

Project Structure:

program_8/

—>nodejs-configmap.yaml

—>nodejs-pod.yaml

—>[server.js](#)

Procedure:

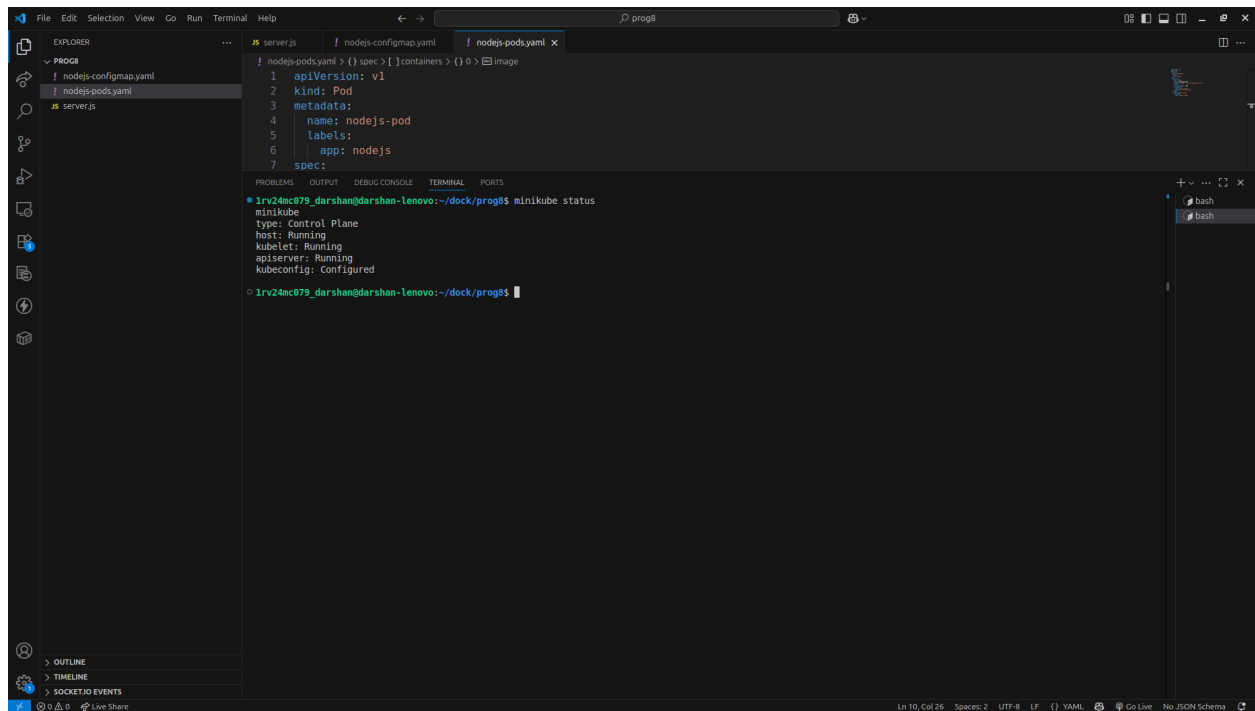
Create a folder (ex: program_8)

Step 1: Start Minikube

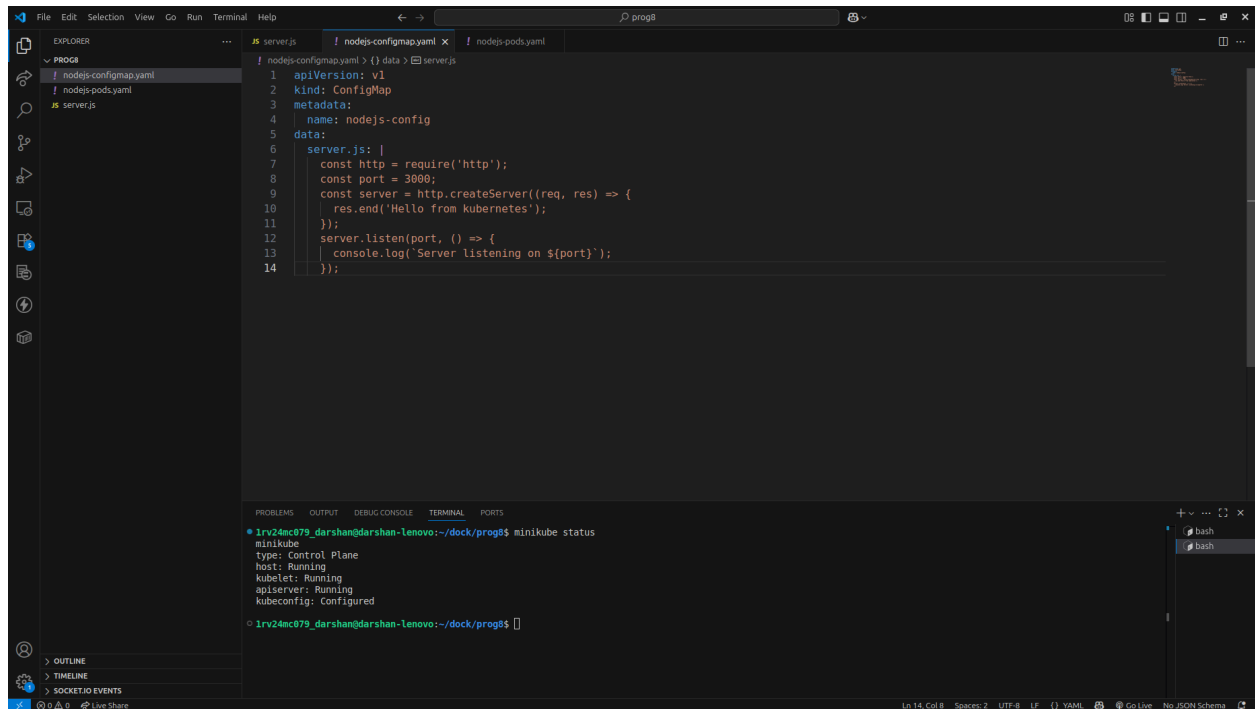
minikube start

Check the status of Minikube

minikube status



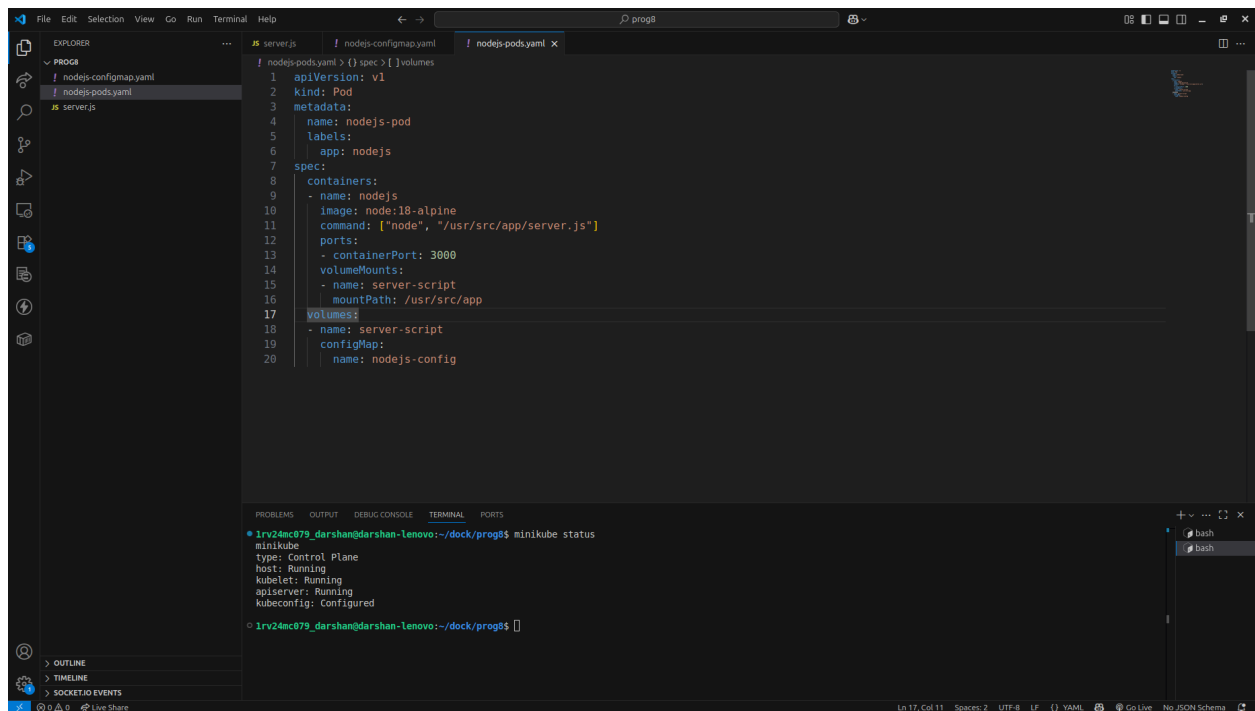
Step 2: Create nodejs-configmap.yaml file with the following code
nano nodejs-configmap.yaml



```
1 apiVersion: v1
2 kind: ConfigMap
3 metadata:
4   name: nodejs-config
5 data:
6   server.js: |
7     const http = require('http');
8     const port = 3000;
9     const server = http.createServer((req, res) => {
10       res.end('Hello from kubernetes');
11     });
12     server.listen(port, () => {
13       console.log('Server listening on ${port}');
14     });
```

```
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

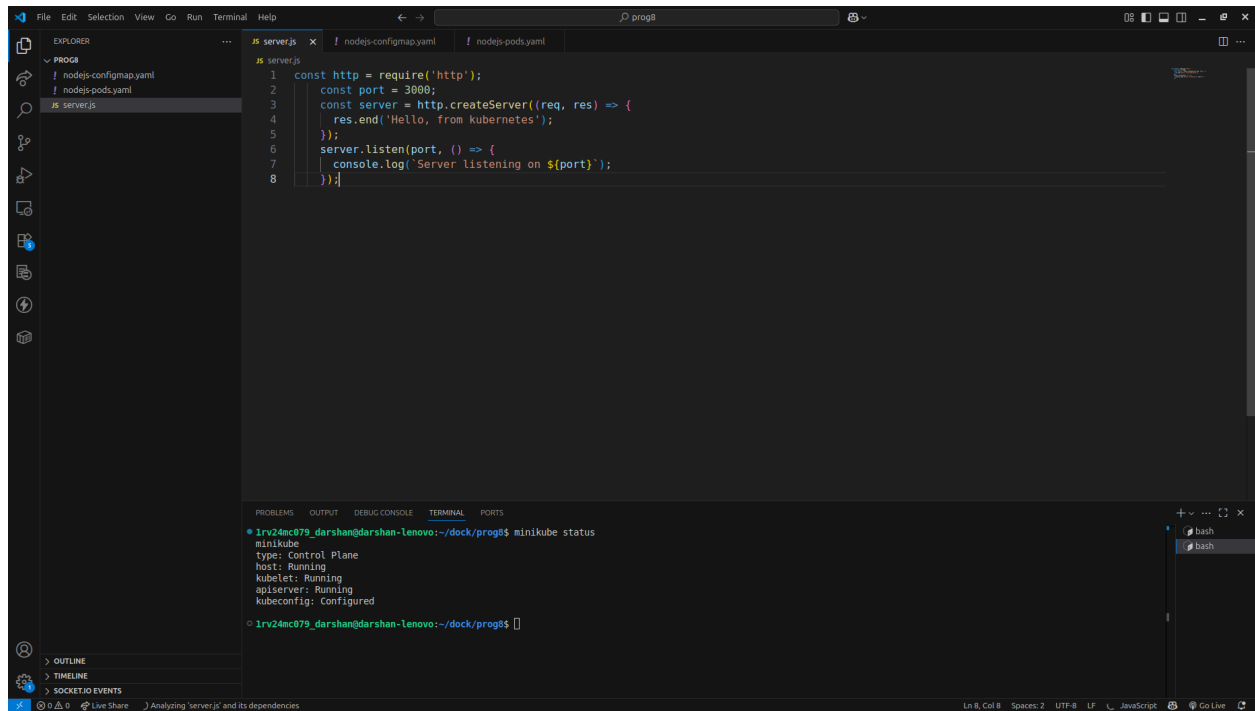
Step 3: Create nodejs-pod.yaml file with the following code
nano nodejs-pod.yaml



```
1 apiVersion: v1
2 kind: Pod
3 metadata:
4   name: nodejs-pod
5   labels:
6     app: nodejs
7 spec:
8   containers:
9     - name: nodejs
10       image: node:18-alpine
11       command: ["node", "/usr/src/app/server.js"]
12       ports:
13         - containerPort: 3000
14       volumeMounts:
15         - name: server-script
16           mountPath: /usr/src/app
17   volumes:
18     - name: server-script
19       configMap:
20         name: nodejs-config
```

```
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 4: Create server.js file with following code
nano [server.js](#)



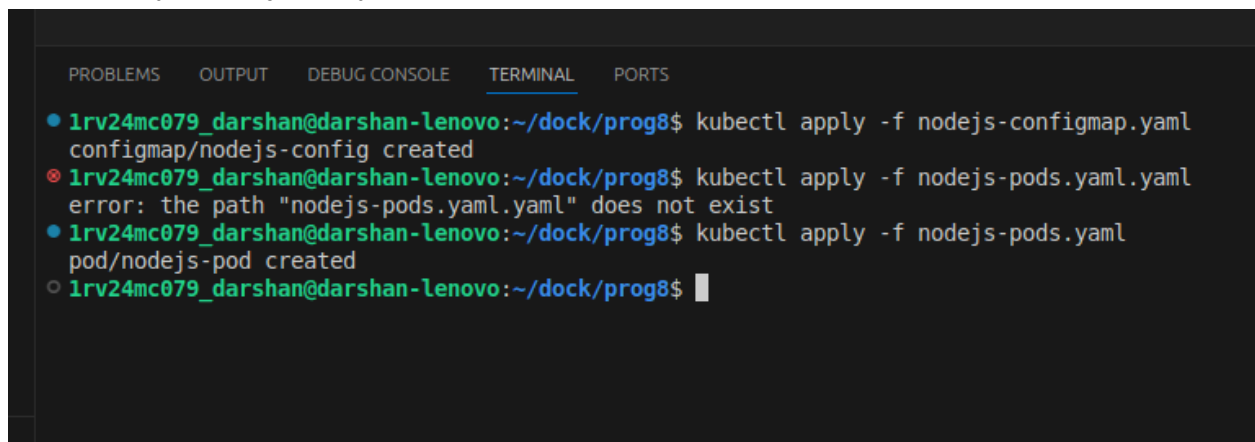
The screenshot shows the VS Code editor interface. The Explorer sidebar on the left shows a project named 'prog8' with files 'nodejs-configmap.yaml', 'nodejs-pods.yaml', and 'server.js'. The main editor window displays the content of 'server.js' with the following code:

```
1 const http = require('http');
2 const port = 3000;
3 const server = http.createServer((req, res) => {
4   res.end('Hello, from Kubernetes');
5 });
6 server.listen(port, () => {
7   console.log('Server listening on %s(port)');
8 });
```

Below the editor, the TERMINAL panel shows the output of the command 'minikube status':

```
1rv24mc079_darshan@darshan-lenovo:~/dock/prog8$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 5: Apply the yaml files
docker apply -f nodejs-configmap.yaml
docker apply -f nodejs-pod.yaml

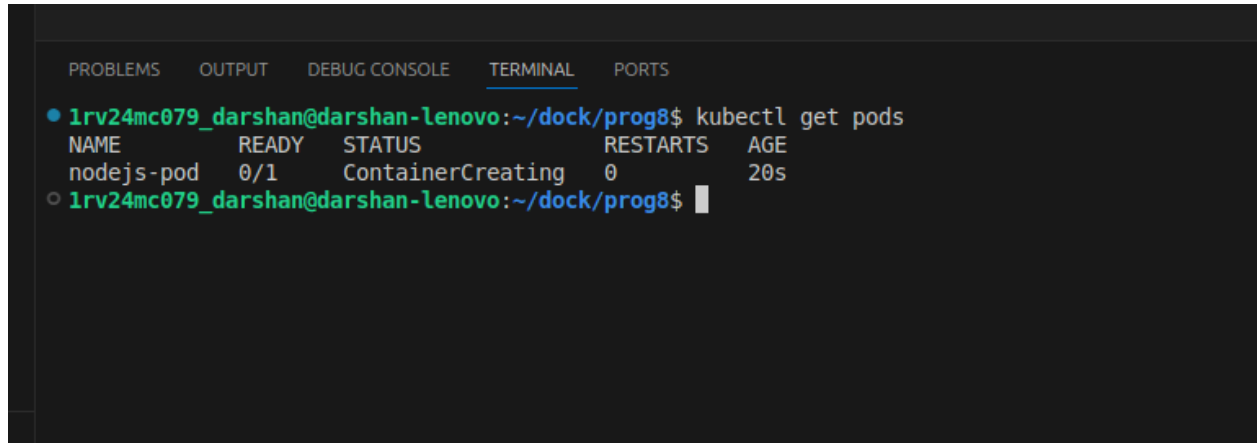


The screenshot shows the terminal output of the following commands:

```
1rv24mc079_darshan@darshan-lenovo:~/dock/prog8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-config created
1rv24mc079_darshan@darshan-lenovo:~/dock/prog8$ kubectl apply -f nodejs-pods.yaml.yaml
error: the path "nodejs-pods.yaml.yaml" does not exist
1rv24mc079_darshan@darshan-lenovo:~/dock/prog8$ kubectl apply -f nodejs-pods.yaml
pod/nodejs-pod created
1rv24mc079_darshan@darshan-lenovo:~/dock/prog8$
```

Step 6: Get the pods

docker get pods



```
1rv24mc079_darshan@darshan-lenovo:~/dock/prog8$ kubectl get pods
NAME          READY   STATUS             RESTARTS   AGE
nodejs-pod    0/1     ContainerCreating   0           20s
1rv24mc079_darshan@darshan-lenovo:~/dock/prog8$
```

Step 7: Access the nodejs application

kubectll port-forward pod/program-8 3000:3000

Step 8: Access on the browser

http://localhost:3000